TRINAY CLINICAL CATARACT REPORT

5-Stage AI Anterior Segment & Lens Opacification Screening

ID: CAT-97885

Date: 29 Sept 2026

PATIENT DEMOGRAPHICS

Patient Name:	**mahi shah**	Age / Gender:	**3 yrs / Female**	Phone:	**09762481967**
Patient ID:	**PX-003**	Examined Eye:	**Right Eye**	Operator:	**avi**

DIAGNOSTIC VERDICT: CATARACT DETECTED

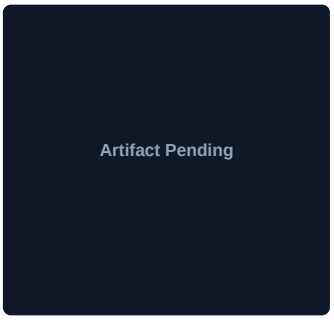
Cataract Prob: 99.9%

Slit-lamp examination & ophthalmologist consultation advised for evaluation.

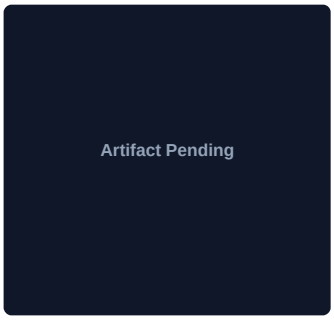
5-STAGE ANTERIOR SEGMENT ANALYSIS BREAKDOWN

1. Image Quality:	**GOOD**	[MobileNetV3-Small]	2. Eye Detection:	**87.6% Confidence**	[YOLO11n Detector]
3. Iris Localization:	**# 248.3 px**	[U-Net Segmenter]	4. Pupil Localization:	**# 149 px**	[U-Net Segmenter]
5. Lens Classifier:	**CATARACT**	[EfficientNet-V2-S + CBAM]	Pupil/Iris Ratio:	**0.600**	[Anatomical Ratio]

VISUAL EVIDENCE GALLERY (PROCESSED ML ARTIFACTS)



1. Original Image



2. Eye Box (YOLO)

3. Iris / Pupil

4. Grad-CAM

CLINICAL RECOMMENDATION & TRIAGE WORKFLOW

Ophthalmologist consultation recommended for slit-lamp examination and cataract staging.

REGULATORY & MEDICAL DISCLAIMER

TRINAY Cataract Screening is an AI-assisted tool intended for primary health triage. Definitive clinical diagnosis, cataract grading, and surgical eligibility must be confirmed by an ophthalmologist.